F47 — WALL 1 KEYSTONE RESOLVED: R(k) uses the K-Casimir ρ_SO(5)=(3/2,1/2) (reproduces recorded values exactly); F46's conformal ρ=(5/2,3/2) is the OTHER, Bergman/Plancherel ρ. Hypothesis A for R(k), inside Hypothesis C (both ρ's real, distinct roles). The two ρ's ARE the bulk/boundary split. Unblocks Walls 2+6.

Lyra (Claude Opus 4.8)

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F47 — Wall 1 Keystone Resolved

0. Result

The F46 c-function wall is resolved, and Casey's diagnosis is exactly right: my naive ρ = (5/2, 3/2) was the conformal ρ; the heat-trace K-type decomposition uses the K-Casimir ρ = ρ_SO(5) = (3/2, 1/2), which reproduces the recorded substrate K-Casimirs exactly:

K-type	K-Casimir, ρ_SO(5)=(3/2,1/2)	recorded
V_(1/2,1/2)	$\|(2,1)\|^2 - 5/2 = 5 - 5/2 = 5/2$	2.5 ⍰
V_(3/2,1/2)	$\|(3,1)\|^2 - 5/2 = 10 - 5/2 = 15/2$	7.5 ⍰
V_(5/2,1/2)	$\|(4,1)\|^2 - 5/2 = 17 - 5/2 = 29/2$	14.5 ⍰

So the F46 "wall" was not an obstruction — it was a convention misassignment between two genuine substrate structures. Hypothesis A holds for R(k) (it uses the K-Casimir), and it sits inside Hypothesis C's truth (both ρ's are real, with distinct roles). Below.

1. The two ρ's of D_IV^5 and their distinct roles

D_IV^5 carries two natural Weyl-vector-type quantities, and they are not interchangeable:

- ρ_SO(5) = (3/2, 1/2) — the half-sum of B_2 positive roots $\{e_1, e_2, e_1-e_2, e_1+e_2\}$ = (3,1)/2. This is the compact Weyl vector. It governs the K-type Casimirs (the discrete K-spectrum), hence the heat-trace K-type decomposition — hence R(k).
- ρ_conformal = (5/2, 3/2) = (n_C/rank, (n_C−2)/rank) — the conformal weight. Note its first component n_C/rank = 5/2 is exactly the Bergman kernel exponent (Ch 5: K_B ∝ (1−zw̄)^{−n_C/rank}). It governs the Bergman kernel / FK Plancherel c-function — the continuous/bulk side.

So: K-Casimir ρ ↔ K-type/heat-trace (discrete); conformal ρ ↔ Bergman/Plancherel (continuous, bulk). R(k) is a heat-trace K-type statement ⍰ it uses ρ_SO(5). I had reached for the conformal ρ because it is the one that appears in the Bergman kernel I work with daily — the misassignment is understandable, and naming it pins the structure.

2. The $2(\lambda_1+\lambda_2)$ offset is the ρ-difference, and it is substrate-natural

The offset between the two conventions (Casey's clue) is exactly the Weyl-vector difference:

$$\rho_{\text{conf}}-\rho_{SO(5)} = (5/2, 3/2)-(3/2, 1/2) = (1, 1), \qquad C_{\text{conf}}(\lambda)-C_{SO(5)}(\lambda) = 2\lambda\cdot(1, 1) = 2(\lambda_1+\lambda_2).$$

For the spinor tower λ=(a,½), this is 2(a+½) = 2,4,6 — Casey's "2(k+1) = rank·(k+1)." The offset is rank · (weight-sum), the linear shift induced by moving between the compact and conformal Weyl vectors. And (1,1) = ((n_C−N_c)/rank, (n_C−2−1)/rank) since n_C−N_c = 2 = rank — substrate-natural. The offset was signal, exactly as Casey said: it is the fingerprint of the two-ρ structure.

3. The C(k,2) source: the quadratic K-Casimir

In the K-Casimir convention, C(j) = $|\lambda+\rho\text{\_SO(5)}|^2 - |\rho\text{\_SO(5)}|^2 = (j+1)^2 - 3/2$ (for the tower, j=1,2,3 → 5/2, 15/2, 29/2). Quadratic in the ρ_SO(5)-shifted weight (j+1). A quadratic spectral variable is the natural source of the pairwise binomial C(k,2) = k(k−1)/2 in R(k): the second-degree structure of the Casimir generates the second elementary-symmetric / pairwise count in the heat-trace coefficients. So the binomial in R(k) = C(k,2)/κ_Bergman descends from the quadratic K-Casimir, and the 1/κ_Bergman = −1/n_C from the Bergman curvature (F41). Both halves now have their source in the correct (K-Casimir) convention.

4. What this closes vs unblocks

Closed (the keystone wall): - The convention is pinned: R(k) uses ρ_SO(5) (K-Casimir), not the conformal ρ. F46's wall dissolves. - The dual-ρ structure is articulated: Hypothesis A for R(k), inside Hypothesis C (both ρ's real; K-Casimir↔discrete/K-type, conformal↔Bergman/Plancherel/bulk). - The C(k,2) source is identified (quadratic K-Casimir).

Unblocked (now well-posed, for Elie/Keeper): - Wall 6 (a_k closed forms): with the convention pinned, the explicit heat-trace coefficients a_k(n_C) in the K-Casimir decomposition can be computed (Elie's parallel support) → the full root-sum = C(k,2)/n_C theorem. F47 supplies the convention; the explicit a_k polynomial is the remaining step. I do not claim the full theorem here — I claim the keystone, and hand the a_k computation to Elie now that it is unblocked. - Wall 2 (F45 N_c^4=81): the question "which H^2 K-type contributes N_c^4 to the muon heat-trace" is now well-posed in the K-Casimir convention — Elie's K-type scan runs against ρ_SO(5) Casimirs, not the conformal ones. (My F45/F46 coupling was right; the convention it needs is now fixed.)

5. The deeper payoff (structural, appropriately tiered)

The two ρ's map onto the bulk/boundary structure that has run through the whole day: - conformal ρ (n_C/rank = Bergman exponent) ↔ the bulk Bergman side; - compact ρ_SO(5) (K-type Casimirs) ↔ the K-type / boundary side.

Under F44 Reading (a), everything physical is in H^2, and the heat-trace K-type decomposition (R(k), and the F45 muon re-derivation) lives on the compact/K-Casimir side. So Wall 1's resolution is *consistent with* Reading (a): R(k) and the muon heat-trace both use ρ_SO(5), the K-type convention. I flag this as a structural observation, not a revived unification claim (I am once burned on that today): it says the conventions line up, not that one operator does everything.

6. Honest status

- RESOLVED (verified): R(k) uses ρ_SO(5)=(3/2,1/2); reproduces recorded K-Casimirs exactly; F46 wall was conformal-vs-compact ρ misassignment.
- ARTICULATED: dual-ρ structure (A within C); offset = rank·(weight-sum) = (1,1) Weyl-difference; C(k,2) from quadratic K-Casimir.
- UNBLOCKED (not claimed): explicit a_k(n_C) → full root-sum theorem (Elie/Wall 6); F45 N_c^4 K-type (Elie/Wall 2) — both now in the pinned convention.
- Tier: F47 v0.1 keystone RESOLVED (convention + structure); full R(k) theorem unblocked, a_k = Elie support; Wall 2 well-posed. @Elie — run the K-type scan and a_k computation against ρ_SO(5) Casimirs (2.5/7.5/14.5), not conformal. @Keeper — Ch 8 absorbs: R(k) = C(k,2)/$\varkappa$_Bergman in the K-Casimir convention; the two-ρ (compact/conformal) structure is

the curvature-vs-spectrum split. @Cal — keystone is a convention pin + structural articulation, not a forced theorem; the theorem awaits a_k.

7. Closure

Wall 1 keystone resolved. R(k) uses the K-Casimir Weyl vector ρ_SO(5) = (3/2,1/2) — reproducing the recorded substrate K-Casimirs exactly — and my F46 conformal ρ = (5/2,3/2) is the genuine *other* ρ governing the Bergman kernel / Plancherel c-function. Both are real (Hypothesis C structure), but R(k), a heat-trace K-type statement, uses the compact one (Hypothesis A verdict). The $2(\lambda_1+\lambda_2)$ = rank·(weight-sum) offset is their Weyl-vector difference (1,1), substrate-natural — signal, as Casey said. The C(k,2) descends from the quadratic K-Casimir. This pins the convention (dissolving the F46 wall), unblocks the explicit a_k computation (Elie/Wall 6 → full root-sum theorem) and the F45 N_c^4 K-type (Elie/Wall 2), and lines up consistently with Reading (a) (K-type/compact side) — stated as structure, not a revived unification. The keystone opened the system, as the wall-attack predicted.

— Lyra, Sat 2026-06-06 14:10 EDT. F47 WALL 1 KEYSTONE RESOLVED: R(k) uses K-Casimir ρ_SO(5)=(3/2,1/2) (half-sum B_2 pos roots) — reproduces recorded substrate K-Casimirs 2.5/7.5/14.5 EXACTLY; F46's ρ=(5/2,3/2) is the CONFORMAL ρ=(n_C/rank,(n_C−2)/rank), the OTHER genuine ρ (governs Bergman kernel exponent n_C/rank=5/2 + FK Plancherel c-function, the bulk/continuous side). Hypothesis A for R(k) (K-Casimir/discrete K-type side), inside Hypothesis C truth (both ρ's real, distinct roles: compact↔K-type/heat-trace, conformal↔Bergman/Plancherel/bulk). Offset C_conf−C_SO5 = $2(\lambda_1+\lambda_2)$ = rank·(weight-sum) = the (1,1)=ρ_conf−ρ_SO5 Weyl-difference (n_C−N_c=2=rank), substrate-natural — Casey's clue was signal. C(k,2) binomial descends from quadratic K-Casimir $C(j)=(j+1)^2-3/2$. CLOSES: convention pinned, F46 wall dissolved. UNBLOCKS (not claimed): explicit a_k(n_C) → full root-sum=C(k,2)/n_C theorem (Elie/Wall 6); F45 N_c^4=81 K-type now well-posed in ρ_SO(5) convention (Elie/Wall 2). Structural: two ρ's = bulk(conformal/Bergman)/boundary(compact/K-type) split, consistent with Reading (a) — stated as structure NOT revived unification (once-burned). @Elie scan against ρ_SO(5) Casimirs not conformal.